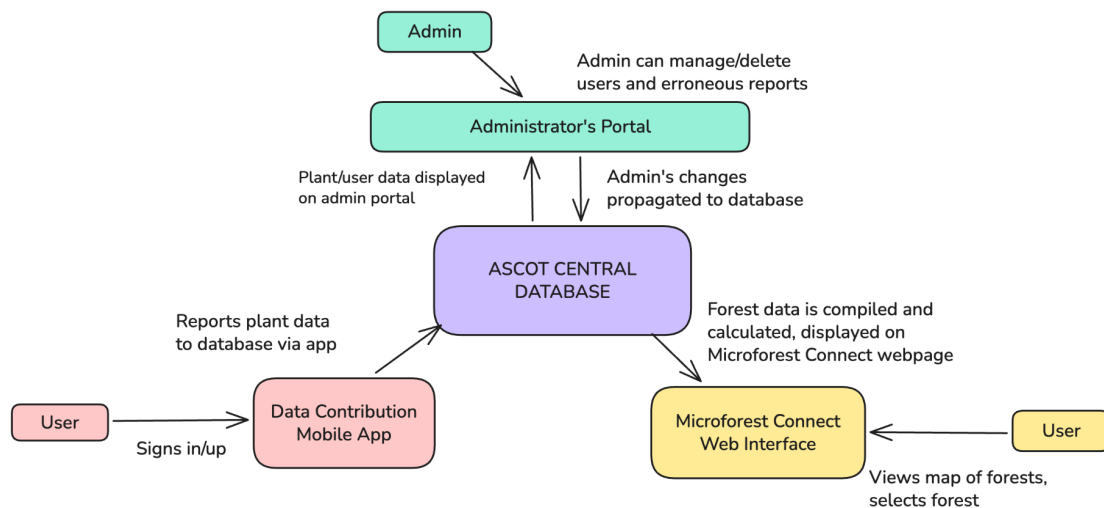


5.0 - Software Requirements Specification

5.1 - Introduction

The Ascot Microforest Connect project is a multi-application system that includes a data contribution interface, an administrator's portal, and an interactive web page, supporting an effort to establish microforests around the Los Angeles area and eventually the world. The complete system will enable microforest visitors and volunteers to report data on the microforests' health, allow the forests' stewards to manage their data and userbase, and provide an informational webpage for the public to learn about the project. Below is a high-level UML diagram of the basic system components and applications, and their interactions with the central Firebase database.



For the rest of this document, each of the following sections has requirements in the form of user stories, that are categorized by which application of the project they apply to. Section 5.2 contains a breakdown of functional requirements that the completed applications will fulfill, section 5.3 contains the specific performance requirements, and finally, section 5.4 contains information detailing the project's necessary environment for both development and execution. Note that these user requirements have newly been converted to the user story format from their

previous “shall” statement format. Also, new user story requirements created due to previous feedback are marked with a **[NEW]** tag for visibility.

5.2 - Functional Requirements

Data Contribution

- **As a field volunteer**, I want to securely log in or sign up for an account, so that my field reports are correctly attributed to my user profile.
- **As a volunteer working outdoors**, I want to use my mobile device's camera to scan a plant's QR code, so that I can enter the Plant ID automatically without making manual typing errors.
- **As a microforest visitor**, I want to fill out a data form detailing a plant's health status, height, and specific observed conditions (like flowering, seeding, or predation), so that I can accurately document the plant's current state for the central database.
- **As a contributor**, I want to view a list of microforests I am already registered with and have a button to register for new forests, so that I can easily submit data to multiple different project sites.

Steward (Administrator) Portal

- **As a microforest ranger**, I want to securely sign in using my email and password (with an option to recover forgotten passwords), so that I can safely access and manage my microforest data without unauthorized interference.
- **As a project manager**, I want to view a dashboard with an Analytics Window displaying headline statistics and data summaries, so that I can quickly assess the overall health and status of a selected forest.
- **[NEW] As a project manager**, I want my steward portal to contain the latest information submitted from the data contribution app reports, so that I have the most up-to-date perspective on the state of my micro-forest.

- **As an administrator/steward**, I want to access a Users Window to view all contributors associated with my forest, so that I can manage user permissions and track who is submitting field data.
- **As a microforest ranger**, I want to view granular plant logs in a Data Window and delete erroneous reports, so that I can clean the collected data and maintain system integrity.
- **As a researcher/administrator**, I want to select a specific forest and download its historical data logs as a CSV file, so that I can perform local data processing, reporting, and external analysis.

Micro-Forest Connect Interface

- **As a public user**, I want to navigate an interactive world map that displays "Microforest Pings", so that I can easily discover the geographic locations of registered microforests.
- **As a community member**, I want to select a specific microforest on the map to view its details sheet (including its title, responsible stewards, and physical scale), so that I can learn more about the organizations managing the local sites.
- **[NEW] As a general viewer**, I want to see the latest compiled forest data and calculated plant health statistics on a public fact sheet, so that I can understand the beneficial environmental impact of these microforests in real time.
- **As a teacher**, I want to access the forest map and quick facts about their impact to show my students that it's easy to get involved on saving the environment!

5.3 - Performance Requirements

Data Contribution

- **As a field volunteer**, I want my plant health logs to submit to the database in under 2 seconds under standard network conditions, so I'm not waiting on loading screens while working outdoors.

- **[NEW] As a field volunteer**, I want clear submission error warnings in the event of network loss or invalid input, so I know to retry or repeat a report submission.
- **As a mobile user**, I want the QR Code Scanner interface to utilize a non-invasive amount of system memory, so that the app doesn't crash during fieldwork.
- **As a registered contributor**, I want my password and personal metadata to be encrypted in the Users Database, so that my private information is protected from unauthorized access.
- **As an outdoor volunteer**, I want the Plant Data Form Tab to be designed for high visibility and ease of use through work gloves and similar hindrances, so that I can easily read and use the app in high-glare, sunny environments.

Steward (Administrator) Portal

- **As a project manager**, I want the dashboard to populate analytics and data summaries within 3 seconds of selecting a forest, so that I can efficiently review information without delay.
- **As an administrator**, I want CSV exports to be optimized so that file sizes do not exceed 50MB for standard monthly logs, so that I can easily download, open, and share reports without dealing with bloated file sizes.
- **As a micro-forest steward**, I want the Forests Database to maintain consistent and replicable data, so that the information I manage on the dashboard perfectly matches what is displayed to the public.
- **As a security-conscious steward**, I want my administrative session to automatically terminate after 30 minutes of inactivity, so that sensitive forest data remains protected if I step away from my computer.
- **As an administrator**, I want a secondary confirmation prompt to appear before I delete a microforest profile or user account, so that I do not accidentally permanently delete critical project records.
- **[NEW] As a project manager**, I want clear error messaging if a CSV download fails or network connectivity is lost, so that I can troubleshoot issues quickly and understand why an action failed.

Micro-Forest Connect Interface

- **As a public user**, I want the map interface to render all microforest pins within 1.5 seconds of loading the page, so that I can quickly start exploring locations without UI lag.
- **As a general viewer**, I want database queries to be heavily indexed so query times do not increase exponentially, so that the website remains fast and responsive regardless of how many thousands of plant logs are added over time.
- **As a curious community member**, I want intuitive drag-and-click navigation on the map with a low learning curve, so that I can easily find microforest details without needing to read a tutorial.

5.4 - Environment Requirements

5.4.1 Development Environment Requirements

- **Frontend**: Development will utilize React Native for the mobile application and React for the web application, styled with Tailwind CSS.
- **Mapping**: The Microforest Connect mapping features will be developed using MapLibre.
- **Backend**: The core backend structure will be developed utilizing Firebase and Node.js.
- **Version Control**: Source code management will be handled using Git and GitHub.
- **Deployment**: The infrastructure stack will use a combination of Firebase hosting, Expo project hosting, and Vercel hosting.
- **Design Tools**: User interface and experience design will be wireframed and modeled using Figma.

5.4.2 Execution Environment Requirements

- **Mobile Hardware**: Requires an 8-core 2.0 GHz or faster processor to handle map rendering and QR scanning. An 8MP or higher camera with auto-focus is required for the QR Code Scanner.
- **Desktop Hardware**: A minimum of 8 GB RAM is required, though 16 GB is recommended for Admin Analytics tasks.

- **Storage:** Devices must have at least 500 MB of available space for the application and local data caching.
- **Connectivity:** 4G LTE, 5G, or Wi-Fi is required to enable real-time database synchronization.
- **App Display:** A high-brightness mobile screen is strongly recommended for successful outdoor use of the Data Contribution app in the field.
- **Desktop Display:** A modern desktop screen of 1280x720px resolution minimum is highly recommended for viewing and use of Micro-Forest Connect and the Steward Portal.
- **Client OS (Mobile):** The application requires Android 10+ or iOS 15+.
- **Client OS (Admin):** Accessible via modern web browsers on Windows 10/11, macOS, or Linux.
- **Web Browser:** Requires Chrome, Firefox, or Safari, and the browser must support WebGL for map rendering.
- **Database System:** The active production environment will run on Firebase.